HUMAN RIGHTS COUNCIL ADVISORY COMMITTEE

Recommended Guidelines for Applying the Existing Human Rights Framework to Neurotechnologies

DRAFT prepared by Vassilis P Tzevelekos, Rapporteur of the drafting group

1. In its resolution A/HRC/RES/58/6, "Neurotechnology and human rights", the Human Rights Council welcomed the previous report of the Advisory Committee on the "Impact, opportunities and challenges of neurotechnology with regard to the promotion and protection of all human rights" (A/HRC/57/61), and requested the Committee "to draft a set of recommended guidelines for applying the existing human rights framework to the conception, design, development, testing, use and deployment of neurotechnologies", for presentation to the Council at its sixty-fourth session. The Committee established a drafting group, currently composed of Ignacio de Casas, Riva Ganguly Das, Alessandra Devulsky, Veronika Fikfak (Chair) and Vassilis Tzevelekos (Rapporteur).
2. Having, in accordance with the Council's request, sought and taken into account the views and inputs of all stakeholders, as well as relevant work already undertaken, including by UNESCO, the Committee adopts, for the purposes of these recommended guidelines, the definitions of "nervous system", "neurotechnology", "neural data" and "whole life cycle" set out in UNESCO's "Recommendation on the Ethics of Neurotechnology" (SHS/BIO/REC-NEURO/2025), and recommends the following guidelines.

Part A

General Framework

1. These recommended guidelines apply across the whole lifecycle of neurotechnologies.

2. These recommended guidelines are underpinned by the understanding that neurotechnologies may constitute both an opportunity for the protection and realization of human rights and a threat to their enjoyment.

   Neurotechnologies may, where developed and used in a manner consistent with applicable international human rights law, contribute notably to health, accessibility, equality of opportunity, inclusion, social participation, autonomy, human well-being, the free development of personality and scientific advancement.

   Neurotechnologies may also pose serious risks to the enjoyment of human rights, notably the rights to life and health, the prohibition of torture and other cruel, inhuman or degrading treatment or punishment, physical and mental integrity, freedoms of thought, opinion and expression, privacy, liberty and security of person, fair trial guarantees, and equality and non-discrimination.

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3. These recommended guidelines reflect a human rights-based approach aimed at preventing and mitigating adverse human rights impacts while enabling the responsible, safe and beneficial governance of neurotechnologies throughout their lifecycle.

4. References in these recommended guidelines to facilitating, promoting or enabling the development, access to, use or deployment of neurotechnologies should be understood as conditional upon consistency with international human rights law and due consideration of the human rights risks involved.

   As part of this approach, appropriate risk assessments should be conducted, drawing on the best available scientific evidence, to identify and assess the potential human rights impacts of neurotechnologies. Such assessments should be comprehensive and take into account, as appropriate, individual, relational and societal dimensions of risk. In assessing the nature and severity of such impacts, attention should be paid to the nature, degree, duration and reversibility of effects on mental as well as physical integrity. The fact that a neurotechnology is physically non-invasive should not, in itself, be regarded as indicating that its interference with human rights is minor.

   Where risks to human rights exist, appropriate preventive measures should be taken. Where, despite such assessment, scientific uncertainty remains regarding the human rights impacts of neurotechnologies, a precautionary approach should be applied. The absence of full scientific certainty should not, in itself, justify refraining from taking measures necessary to prevent or mitigate serious, irreversible or widespread adverse human rights impacts.

5. Recognizing that neurotechnologies remain rapidly evolving and that their applications, effects and implications for human rights may not yet be fully identified, understood or assessed, these recommended guidelines should be read in light of scientific developments, evolving knowledge, continuous application by international human rights institutions of international human rights law to neurotechnologies, and emerging human rights opportunities and challenges.

   The continued relevance of these recommended guidelines should be kept under review.

6. These recommended guidelines are grounded in the existing international human rights framework. Throughout these recommended guidelines, references to international human rights law should be read together with relevant international human rights standards and principles, including human dignity, autonomy and equality.

7. The use of "should" throughout these recommended guidelines reflects their recommendatory character and is without prejudice to any applicable international legal obligations.

8. These recommended guidelines reflect:

   a) States' existing international human rights obligations;

   b) The role and influence of business enterprises in relation to neurotechnologies across their lifecycle, and, consistent with the United Nations Guiding Principles on Business and Human Rights, the responsibility of such enterprises to conduct their activities in

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conformity with internationally recognized human rights rules and address adverse impacts;

c) The role and responsibility of other relevant actors, within their respective functions and powers, to conduct their activities relating to neurotechnologies across their lifecycle in conformity with internationally recognized human rights rules and, as appropriate, to contribute through their practice to the application, interpretation, development and oversight of such rules.

9. Where these recommended guidelines are observed, they should be applied in a non-discriminatory manner to effectively safeguard the rights of all persons affected by neurotechnologies, with particular attention to individuals and groups who are marginalized or in situations of vulnerability, or at risk of marginalization or heightened vulnerability, including children, persons with disabilities, older persons, persons in situations of *de jure* or *de facto* deprivation of liberty by public authorities or private entities, racial and ethnic groups, including people of African descent, Indigenous Peoples, workers, migrant workers, military or security personnel, and persons exposed to dependency relationships, structural discrimination, significant imbalances of power or other situations that may impair autonomy, increase susceptibility to coercion or otherwise heighten the risk of adverse human rights impacts.

   Particular attention should be paid to preventing and addressing risks of direct, indirect and intersectional discrimination arising from neurotechnologies.

   Meaningful consultation, engagement with and active involvement of affected persons and groups should be sought, particularly where neurotechnologies may significantly affect their rights.

10. Recognizing that access to neurotechnologies may be significantly unequal within and between States, these recommended guidelines reflect the need for the development and diffusion of neurotechnologies, where they offer benefits, to contribute, as far as possible, to reducing rather than exacerbating existing inequalities.

Part B

Roles and Responsibilities of Relevant Actors

States

11. Consistent with their obligations under international human rights law, States should respect, protect and fulfill human rights throughout the lifecycle of neurotechnologies.

12. States should refrain from interfering with the freedom, self-determination and autonomy of individuals to use and otherwise engage with neurotechnologies.

    However, to protect human rights, States may be required to adopt measures limiting such freedom, self-determination and autonomy.

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13. In no circumstances whatsoever may States employ neurotechnologies in a manner incompatible with absolute human rights.

14. Any State interference with qualified human rights through the use of neurotechnologies, or through measures limiting individuals' freedom to use or otherwise engage with neurotechnologies, should have a clear basis in law, pursue a legitimate purpose and comply with the principle of proportionality, including the requirement of necessity.

15. States should, within their territory and/or jurisdiction, protect against human rights violations by any actor involved in any stage of the lifecycle of neurotechnologies, including business enterprises.

    To this end, States should employ all appropriate means within their capacities, including through policies, legislation, regulation, monitoring, enforcement and adjudication. In particular, States should seek to prevent human rights violations, including by identifying, assessing and mitigating risks and adverse human rights impacts associated with neurotechnologies; stop ongoing violations; promptly and effectively investigate alleged violations and, where appropriate, impose sanctions; provide avenues for accountability, including through access to effective judicial and non-judicial mechanisms capable of offering remedies to victims.

16. Measures to prevent adverse human rights impacts may include, where warranted by the nature and severity of the risks involved, restrictions or prohibitions on particular uses or applications of neurotechnologies.

    Where required under international law, States should go beyond prohibiting conduct amounting to breaches of international human rights law by criminalizing the most serious such breaches committed by actors involved in any stage of the lifecycle of neurotechnologies falling within their territory and/or jurisdiction.

17. Considering in particular the role and influence of business enterprises in relation to neurotechnologies across their lifecycle, States should set and enforce rules requiring business enterprises within their jurisdiction, including, where applicable, with respect to their operations outside the State's territory, to conduct business activities in conformity with internationally recognized human rights rules.

18. States should conduct human rights due diligence before procuring, financing or deploying neurotechnologies developed by any actor, including business enterprises, academic and research institutions or public-private consortia, and throughout the period of their involvement with such technologies.

19. States should establish appropriate controls over the export, transfer or sale of neurotechnologies where there is a risk that they may be used in ways incompatible with international human rights law or, where applicable, international humanitarian law, including, where necessary, by restricting or prohibiting such transactions.

20. In light of the evolving nature of neurotechnologies, States should keep under regular review the adequacy and effectiveness of legal and regulatory frameworks and take proactive steps to identify and address gaps in these frameworks and in their implementation and enforcement.

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21. To support the safe development and realization of the potential benefits of neurotechnologies, States should, as appropriate, enable controlled and regulated environments for their development, experimentation and testing, subject to effective independent oversight and appropriate human rights and ethical safeguards, including those relating to medical ethics.

22. Where neurotechnologies offer benefits, States should facilitate safe and equitable access to neurotechnologies through inclusive approaches responsive to the diverse needs of persons within their jurisdiction.

23. Where States make use of neurotechnologies in providing services or other forms of support to persons within their jurisdiction, including in contexts such as healthcare, they should do so in a non-discriminatory manner and, where necessary, adopt affirmative measures to promote substantive equality.

24. States should cooperate with one another and with relevant international and regional organizations in safeguarding human rights throughout the lifecycle of neurotechnologies. In pursuing such cooperation, States should engage in activities such as capacity-building, technology transfer on fair and mutually agreed terms, scientific collaboration, open and collaborative approaches to innovation, and the sharing of knowledge and good practices.

25. In engaging in international cooperation, States should seek to promote equitable access to the benefits of neurotechnologies and reduce economic, scientific, technological and regulatory disparities between States.

    States should also seek to ensure that the development, commercialization, deployment and use of neurotechnologies do not exacerbate existing inequalities or contribute to new forms of exclusion, dependency or concentration of technological, economic or political power, whether within or between States.

Business Enterprises

26. Business enterprises involved in any stage of the lifecycle of neurotechnologies, including development, production, commercialization, deployment or use, should respect internationally recognized human rights throughout their operations and abstain from causing or contributing to adverse human rights impacts.

27. Business enterprises should adopt and implement appropriate human rights policies relating to neurotechnologies and integrate respect for internationally recognized human rights throughout their governance structures, decision-making processes and business relationships.

28. Business enterprises should implement human rights due diligence through proactive measures within their capacities to identify, prevent and mitigate adverse human rights impacts that they may cause or contribute to, or that may be directly linked to their operations, products or services by their business relationships, and to account for how such impacts are addressed.

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Such due diligence should be appropriate to the nature and context of their operations and to the severity and likelihood of the risks posed by neurotechnologies.

29. Where business enterprises cause or contribute to an ongoing adverse human rights impact, they should cease the impact or their contribution to it, as appropriate.

    Where they have caused or contributed to adverse human rights impacts, they should provide for or cooperate in their remediation.

30. Business enterprises should communicate in a transparent manner concerning the human rights risks associated with their neurotechnology-related operations, products and services and the measures adopted to address them.

31. Where appropriate, business enterprises should engage in meaningful consultation with affected stakeholders, including persons and groups who are marginalized, in situations of vulnerability or potentially disproportionately affected by neurotechnologies.

32. Where applicable domestic law provides for higher standards of human rights protection, business enterprises involved in any stage of the lifecycle of neurotechnologies should comply with such standards.

33. The responsibilities of business enterprises involved in any stage of the lifecycle of neurotechnologies apply independently of the ability or willingness of States to comply with their human rights obligations.

34. Business enterprises should not exploit regulatory gaps, weak enforcement environments or differences between domestic legal systems to evade their human rights responsibilities.

Other Relevant Actors

35. Other relevant actors involved in any stage of the lifecycle of neurotechnologies should conduct their activities in conformity with applicable internationally recognized human rights rules. As relevant, they should take proactive measures within their respective functions and powers to identify, prevent and mitigate adverse human rights impacts and, where such impacts are ongoing, seek to cease them.

36. Relevant international and regional intergovernmental organizations have an important role to play, within their respective competences and mandates, in contributing to the application, interpretation, development and oversight of international human rights law in relation to neurotechnologies, including through cooperation, technical assistance, standard-setting, monitoring and adjudicatory or quasi-adjudicatory mechanisms.

    The performance of such roles should be actively pursued and enabled.

Part C

Cross-Cutting Governance and Human Rights Safeguards

Consent

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37. In light of the risks that neurotechnologies may pose to human rights, States should, as a general rule, require that activities involving the use of neurotechnologies by or on individuals, or their application in any other manner that affects them, be subject to their free, prior, informed, real, genuine, transparent, effective and explicit consent.

    Throughout these recommended guidelines, references to consent should be understood as requiring that consent possess these characteristics.

    The requirement of consent applies subject to any qualifications permissible under international human rights law.

38. Consent should be based on information that is clear and comprehensible, provided, where appropriate, in accessible formats and in simplified form, and should cover the nature, purpose and risks of the use or other application of neurotechnologies, including possible misuse, side effects and unintended consequences.

39. Consent should, as a general rule, be capable of being withdrawn at any stage, without unjustified adverse consequences.

    Any restrictions on the ability to withdraw consent should pursue a legitimate protective purpose, remain consistent with international human rights law and be subject to appropriate safeguards, oversight and effective remedies.

40. Consent should not be regarded as valid where it is obtained through coercion or undue influence, or where it does not possess the requisite characteristics.

    States should exercise particular caution in situations involving deprivation of liberty, institutionalization, dependency relationships, educational authority structures, age, institutional pressures or significant imbalances of power, where the conditions necessary for consent may be compromised.

41. Consent should not be regarded as valid where it is made a condition for access to services, opportunities or benefits in circumstances where individuals do not have a reasonable alternative or cannot refuse without suffering significant adverse consequences.

    States should seek to prevent reliance on such conditional consent, particularly where individuals may be compelled, directly or indirectly, to accept the use or other application of neurotechnologies to access essential services or participate in social or economic life.

    This is without prejudice to any use or application without consent that is permissible under international human rights law.

42. Consent should not, in itself, be regarded as sufficient to justify the use or other application of neurotechnologies where the risks to human rights require additional safeguards or restrictions. In such cases, States should regulate accordingly to ensure consistency with international human rights law.

43. Certain uses or applications of neurotechnologies that are incompatible with human dignity or international human rights law cannot be rendered permissible by consent. States should prohibit such uses or applications.

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44. In relation to children, States should take into account their age and evolving capacities in determining their ability to provide consent. Where a child is unable to provide consent, any consent provided on the child's behalf should be consistent with the best interests of the child and international human rights law. In all cases, the views of the child should be given due weight in accordance with the child's age and maturity.

45. In relation to persons who may require support in providing consent, States should seek to ensure respect for their legal capacity on an equal basis with others and provide access to appropriate support for decision-making. Such support should respect their rights, will and preferences and be subject to appropriate safeguards. Where, despite significant efforts and support, a person's will and preferences cannot be determined in practice, any decision should be based on the best interpretation of their will and preferences and comply with international human rights law.

Testing and Experimentation Involving Human Participants

46. Testing and experimentation involving human participants in the development and assessment of neurotechnologies, whether for medical or non-medical purposes, should be conducted in accordance with international human rights law and relevant ethical standards, including, where applicable, medical ethics, as well as requirements relating to consent and independent and impartial oversight.

Unauthorized Access, Misuse and Manipulation

47. States should seek to prevent and address risks arising from unauthorized access to, manipulation of, or control over neurotechnologies, including where these occur through cybersecurity breaches or other cyber-enabled means.

48. States should prohibit the use of neurotechnologies for purposes of social control or coercive behavioral conformity based on individuals' personal thoughts, beliefs, convictions, opinions, preferences, or other aspects of their identity or status.

49. States should prohibit the use of neurotechnologies to manipulate or behaviorally condition individuals' thoughts, beliefs, convictions, opinions, preferences, emotions or decisions, including for commercial, marketing, political or other purposes, when they are in states or circumstances in which their capacity to exercise meaningful autonomy is impaired or they are particularly susceptible to influence, such as during sleep or dreaming.

Neurotechnology-Related Data and Information

50. States should regulate, in a manner consistent with international human rights law, neurotechnology-related data and information. For the purposes of these recommended guidelines, this term encompasses neural data; non-neural data, including personal, biometric, behavioral, communications, geolocation or other data, collected, generated, processed or otherwise used in connection with neurotechnologies; and information derived, inferred or interpreted from neural or non-neural data pertaining to neural activity, cognitive processes or mental states.

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51. States should seek to prevent and address human rights risks arising from the use of neurotechnology-related data and information.

    Such measures should, in particular, respond to risks arising from manipulation, behavioral profiling, exploitative practices, discriminatory treatment or undue influence, especially where asymmetries of power, structural inequalities, situations of dependency or vulnerability are present or where artificial intelligence is used to process or interpret neurotechnology-related data and information or to derive inferences from them.

    States should pay particular attention to the long-term societal implications of the aggregation, commercialization and use of such data and information, including risks to equality, social inclusion, civic engagement and participation, as well as risks of manipulation and concentration of economic, technological or political power.

52. States should seek to ensure that neurotechnology-related data and information, including inferences derived from them, are not treated as inherently objective or determinative of individuals' abilities, preferences, behavior, identity or capacity to make decisions, or used in ways that result in discrimination or unjustified restrictions on the exercise of human rights.

    Where such data and information are used in decision-making that significantly affects individuals, States should seek to ensure appropriate transparency concerning the basis of such decisions and meaningful opportunities to challenge them.

    States should also seek to ensure that neurotechnology-related data and information used in the development, training, testing or validation of neurotechnologies are sufficiently representative of the populations and contexts in which such neurotechnologies are intended to be used, taking into account relevant cultural and contextual differences.

53. States should recognize the particularly sensitive nature of neural data, as well as other neurotechnology-related data and information capable of revealing or supporting inferences concerning mental states, cognitive processes or neurophysiological functioning and seek to ensure that they are subject to enhanced protection and safeguards, including with regard to their collection, processing, storage, transfer, commercialization, reuse, sharing and other forms of use across the lifecycle of neurotechnologies.

    Such safeguards should include, as applicable, purpose limitation, data minimization, data accuracy and quality, limitations on retention, mechanisms for access and rectification and, where appropriate, deletion, security, transparency, consent, independent oversight and effective remedies.

    Transborder transfers should be subject to appropriate safeguards ensuring continuity of protection.

54. States should, in particular, regulate the commercialization, transfer and use of neurotechnology-related data and information with a view to preventing exploitation, abuse, excessive concentration of power or other adverse human rights impacts.

    Where warranted by the nature and severity of the risks involved, States may consider adopting heightened restrictions that could go as far as the non-commodification of certain

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such data and information, including through restrictions or prohibitions on their commercialization or transfer.

55. States should promote the responsible and rights-consistent access to, use and sharing of neurotechnology-related data and information for healthcare, scientific research and other legitimate public-interest purposes, subject to appropriate safeguards relating to privacy, confidentiality, security, consent, anonymization or pseudonymization, as applicable, and protection against use for incompatible purposes, re-identification or unauthorized access.

    States should also promote the equitable sharing of the benefits arising from such use and research.

56. States should establish appropriate frameworks and mechanisms to ensure the responsible stewardship, accountability and oversight of neurotechnology-related data and information, including through independent and interdisciplinary supervision, human rights impact assessments, transparency requirements and effective review and redress mechanisms.

Commercialization for Consumer Use

57. States should regulate the commercialization of neurotechnologies for consumer use with a view to preventing and mitigating adverse human rights impacts, ensuring safety and safeguarding consumer rights. Such regulation should also determine whether such neurotechnologies may be commercialized and, if so, under what conditions, including, where necessary, requirements for professional supervision or restricted access.

58. States should seek to ensure that accessible information is provided concerning the scientific basis, functions, effects, limitations and potential risks associated with neurotechnologies commercialized for consumer use. Such measures should include, as appropriate, independent oversight, transparency, labeling and marketing requirements.

59. States should monitor on an ongoing basis the effects and risks associated with neurotechnologies commercialized for consumer use and, where necessary, adopt appropriate regulatory and enforcement measures to prevent and mitigate adverse human rights impacts.

60. States should regulate the commercialization of neurotechnologies authorized or certified for medical purposes, including through appropriate safeguards relating to safety, oversight, transparency, accessibility and user protection, and by determining whether and under what conditions such neurotechnologies may appropriately be commercialized for consumer use or should remain subject to professional supervision or restricted access.

61. States should regulate neurotechnologies marketed or presented as producing medical, therapeutic, health-related or performance-optimization effects that are not medically authorized or certified, including by determining whether and under what conditions such neurotechnologies may be commercialized for consumer use or should remain subject to professional supervision or restricted access.

62. States should seek to ensure that neurotechnologies marketed or presented as producing medical, therapeutic, health-related or performance-optimization effects that are not

medically authorized or certified are clearly identified and labeled as such and do not create misleading impressions concerning their scientific basis, functions or effects.

63. States should regulate the use of neurotechnologies and neurotechnology-related data and information for commercial or marketing purposes where they are capable of influencing or manipulating individuals' mental processes, emotions, decision-making or behavior. Such regulation should seek to prevent and address practices that are coercive, deceptive, exploitative or otherwise incompatible with autonomy, freedom of thought or opinion, mental integrity, privacy or other applicable human rights.

Human Augmentation and Cognitive Enhancement

64. States should seek to distinguish between neurotechnologies intended for medical, therapeutic or assistive purposes and those intended for human augmentation or cognitive enhancement. Such distinction should take into account, as appropriate, the purpose, context, claimed effects, manner of use, scientific basis and risks associated with the neurotechnology concerned.

    States should also recognize that the same neurotechnology may, depending on the context, purpose or individual concerned, constitute a therapeutic or assistive application in one case and an augmentation or enhancement application in another.

65. States should seek to restrict or, where necessary, prohibit applications of neurotechnologies for human augmentation or cognitive enhancement that create serious risks to human rights, taking particular account of their non-therapeutic nature and the extent and reversibility of the alteration involved. Such risks may include coercion, subjugation, exploitation, discrimination, manipulation, serious or irreversible effects, or uses incompatible with human dignity or international human rights law.

    States should exercise particular caution in military, security, employment or educational settings, or in settings of *de jure* or *de facto* deprivation of liberty.

66. States should prohibit individuals from being compelled, directly or indirectly, to use neurotechnologies for human augmentation or cognitive enhancement purposes, including as a condition for access to employment, education, insurance, healthcare, services, benefits, opportunities or social participation, or for the purposes of military, security or law enforcement activities or service.

67. Any departure from this prohibition should be confined to exceptional circumstances where such compulsion is compliant with international human rights law and the relevant use or application is subject to appropriate safeguards, oversight and effective remedies.

    States should pay particular attention to situations involving structural inequalities, dependency relationships or significant imbalances of power.

68. States should regulate neurotechnologies intended for human augmentation or cognitive enhancement with a view to preventing and mitigating adverse human rights impacts, including risks to human dignity, autonomy, freedom of thought, mental integrity, privacy, equality and non-discrimination. Such regulation should include, as appropriate, scientific

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assessment, independent and impartial oversight, human rights impact assessments and ongoing review mechanisms.

States should exercise particular vigilance in light of the evolving, uncertain or potentially irreversible effects and human rights implications associated with certain forms of human augmentation or cognitive enhancement, particularly those aimed at substantially altering human capacities or characteristics, or capable of having broader societal implications.

69. States should foster interdisciplinary, independent and evidence-based research on neurotechnologies intended for human augmentation or cognitive enhancement, including their short-, medium- and long-term impacts on equality, autonomy, civic participation, labor relations and social structures.

70. States should regulate the commercialization of neurotechnologies intended for human augmentation or cognitive enhancement purposes, including through appropriate requirements relating to safety, scientific evidence, transparency, labeling, marketing, accessibility of information and independent oversight. States should seek to prevent misleading or unsubstantiated claims concerning the functions, effects, scientific basis or benefits of such neurotechnologies.

71. States should seek to prevent the use of neurotechnologies for human augmentation or cognitive enhancement purposes from contributing to discrimination, exclusion, exploitation, social stratification or new forms of inequality that undermine equality of opportunity or equitable participation in social, economic, educational or public life, including inequalities linked to economic or social status, disability, access to technology or other protected characteristics.

    States should also take measures to seek to mitigate inequalities and structural disadvantages within and between States associated with technological divides or unequal access to such neurotechnologies, including by promoting accessibility and reducing exclusionary barriers where appropriate.

Dependency Relationships and Imbalances of Power

72. States should exercise heightened vigilance and adopt appropriate safeguards with regard to the development, commercialization, deployment and use of neurotechnologies in contexts characterized by dependency relationships, significant imbalances of power, heightened vulnerability or particular susceptibility to coercion or undue influence, including in intimate, familial, caregiving, institutional, employment, military, security, educational or deprivation of liberty contexts.

    States should pay particular attention to risks of coercion, manipulation, surveillance, abuse, exploitation, discrimination or undue influence, including in the private sphere.

73. States should also seek to ensure that where neurotechnologies are employed in contexts of dependency or significant imbalances of power for legitimate purposes such as safety, health, the prevention of harm or the protection of affected persons or third parties, such uses remain consistent with international human rights law and are subject to appropriate safeguards, oversight and effective remedies.

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Part D

Neurotechnologies in Specific Contexts

Medical

74. In pursuing the effective enjoyment of the highest attainable standard of physical and mental health, States should take all appropriate measures within their capacities to facilitate access to safe and human rights-compliant neurotechnologies for medical or assistive purposes, where these can contribute to the prevention, diagnosis or treatment of health conditions, rehabilitation or other health-related needs.

75. In the use of neurotechnologies for medical or assistive purposes, States should seek to ensure that such uses remain consistent with international human rights law and are subject to appropriate safeguards, oversight and effective remedies. States should pay particular attention to respect for autonomy, equality and non-discrimination, and to the meaningful participation of and consultation with affected persons.

76. States should promote accessibility and inclusion and seek to ensure non-discrimination in relation to safe and human rights-compliant neurotechnologies intended for medical or assistive purposes.

77. States should afford particular safeguards and protection to persons in situations of vulnerability and persons exposed to dependency relationships or significant imbalances of power in relation both to decisions concerning the medical or assistive use of neurotechnologies and to their use in practice, including children, older persons, persons with disabilities, persons with physical or mental health conditions, and other persons in situations of heightened vulnerability.

78. States should seek to ensure that neurotechnologies intended for medical purposes are subject throughout their lifecycle to appropriate testing, scientific assessment, certification, independent oversight, regulatory supervision, continuous monitoring and review, and appropriate arrangements for maintenance, support, updates and discontinuation, supported by scientific evidence concerning their safety, efficacy, risks and adverse human rights impacts.

    Such arrangements should seek to ensure continuity of necessary care and support and, where appropriate, continued access to the neurotechnology or suitable alternatives where a neurotechnology is discontinued or ceases to be supported, including as a result of commercial withdrawal, insolvency or other changes affecting its availability.

    States should also seek to ensure that identified risks are effectively prevented or mitigated.

79. States should seek to ensure that medical and other health professionals involved in recommending, prescribing, administering, monitoring or supervising the use of neurotechnologies act in a manner consistent with international human rights law and applicable standards of medical ethics. In particular, they should respect individuals' autonomy, will and preferences in decisions concerning their healthcare; comply with the requirements concerning consent; provide relevant information in accessible and comprehensible forms; protect privacy and confidentiality; and facilitate the meaningful

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participation and consultation of affected persons in decisions concerning the use of neurotechnologies.

Education and Learning

80. States should exercise particular caution regarding the development, deployment and use of neurotechnologies in educational, training and learning contexts at all stages of life.

    Particular caution is warranted regarding practices involving profiling, monitoring, behavioral tracking, inference or assessment of attention, concentration, emotions, stress, behavior, cognitive performance, engagement, aptitude, intentions or other mental or neural processes, particularly where such practices may significantly affect educational opportunities, disciplinary processes, evaluation, ranking, access to opportunities or future life prospects.

    Particular caution is also warranted regarding the use of neurotechnologies to influence, modify or enhance learners' cognition, emotions, behavior or performance, including for purposes of educational optimization, behavioral influence, cognitive or emotional modulation, performance enhancement or similar purposes.

81. States should prohibit the deployment and use of neurotechnologies, and the use of neurotechnology-related data and information in educational, training and learning contexts in ways incompatible with international human rights law, including through manipulative or discriminatory practices.

82. States should seek to ensure that the use of neurotechnologies in educational, training and learning contexts does not undermine autonomy, independent judgment, decision-making, critical thinking, creativity, the freedoms of thought and expression, including academic freedom, mental integrity, the free development of personality or intellectual development.

    Such risks may arise, among other ways, through behavioral conditioning, cognitive interference, or chilling effects on thought or expression.

83. States should seek to ensure that learners, students or other persons in educational, training or learning contexts are not compelled, directly or indirectly, or subjected to undue pressure or incentives to use neurotechnologies or otherwise be subject to their application, including in order to obtain educational advantages, improve academic performance, remain competitive or gain access to educational opportunities, except to the extent that any such measure is compliant with international human rights law and the relevant use or application is subject to appropriate safeguards, oversight and effective remedies.

84. Where, in pursuing the effective enjoyment of the right to education, States facilitate or otherwise support the deployment or use of neurotechnologies in educational, training and learning contexts for purposes relating to accessibility, inclusion, equality, communication, assistive and individualized learning support, meaningful participation in educational activities, the enhancement of learning experiences, safety, the prevention of serious harm or the protection of learners, staff or third parties, and other legitimate educational or supportive purposes, they should do so in conformity with international human rights law and subject to appropriate safeguards, oversight and effective remedies.

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Where neurotechnologies are employed for assistive, accessibility or inclusion purposes, their use should respect autonomy, dignity, neurodiversity and the right of learners to participate meaningfully in decisions affecting them.

85. States should pay particular attention to preventing and addressing the risks of exacerbating inequality, discrimination, exclusion, stigmatization, segregation, unequal educational opportunities and pressures to conform to particular cognitive, behavioral or neurological characteristics arising from the use or non-use of neurotechnologies, including from unequal access to neurotechnologies, neurotechnological profiling, behavioral prediction, or the use of neurotechnology-related data and information for evaluative, predictive or enhancement purposes.

86. States should pay particular attention to the long-term societal implications of intrusive neurotechnological monitoring or behavioral tracking practices, cognitive profiling and enhancement practices in educational contexts, including risks relating to inequality, exclusion, dependency, concentration of power, commercialization of education and erosion of human autonomy and civic participation.
87. States should subject the use of neurotechnologies in relation to children in educational, training and learning contexts to heightened scrutiny and appropriate safeguards. In all actions concerning children, the best interests of the child should be a guiding principle and a primary consideration.

    States should pay particular attention to children's age, evolving capacities, neural and psychological development, heightened susceptibility to coercion, manipulation, behavioral conditioning and peer or institutional pressure, and to the possible long-term or irreversible effects of neurotechnologies on cognition, emotional development, personality, autonomy or independent judgment.

    States should seek to ensure that neurotechnologies are not used in educational contexts in ways that undermine children's dignity, autonomy, creativity, emotional development, freedom of thought or ability to develop independent views and identities.

Employment and Workplace

88. States should exercise particular caution regarding the development, deployment and use of neurotechnologies in employment and workplace contexts, in light of the significant imbalances of power and dependency relationships that may arise in such contexts.

    Particular caution is warranted regarding practices involving the monitoring, inference or assessment of attention, emotions, stress, fatigue, cognitive performance, behavior, intentions or other mental or neural processes of workers or other persons in workplace contexts, particularly where such practices may affect recruitment, continued employment, promotion, performance evaluation, remuneration, benefits, access to opportunities or other work-related interests.

    Particular caution is also warranted regarding the use of neurotechnologies for purposes of productivity optimization, behavioral influence, cognitive or emotional modulation, performance enhancement or similar purposes.

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89. States should seek to prevent and address adverse human rights impacts arising from the use of neurotechnologies and neurotechnology-related data and information in employment and workplace contexts, including discriminatory treatment, exclusion, stigmatization, profiling or unequal treatment, including on the basis of disability, physical or mental health conditions, neurodiversity or other protected characteristics, particularly where such impacts arise from probabilistic inferences, behavioral profiling, automated decision-making or unequal social or structural conditions.

    States should also seek to prevent and address uses of neurotechnologies that undermine autonomy, independent judgment, freedoms of thought and expression, mental or physical integrity, privacy, civic participation or the enjoyment of other applicable human rights, including through behavioral conditioning, cognitive interference or chilling effects on thought, expression or participation.

90. States should prohibit the deployment or use of neurotechnologies, and the use of neurotechnology-related data and information, in employment and workplace contexts in ways incompatible with international human rights law, including through exploitative, manipulative or discriminatory practices or through unlawful surveillance.

91. Where neurotechnologies are deployed or used in employment and workplace contexts for legitimate purposes, including occupational safety, health, accessibility, inclusion, assistive support, the facilitation of meaningful participation in employment or the prevention of harm to workers or third parties, States should seek to ensure that such deployment or use is consistent with international human rights law, including just and favorable conditions of work, and subject to appropriate safeguards, oversight and effective remedies.

    Such safeguards should include, as appropriate, transparency requirements, independent supervision, human rights impact assessments, accessible complaints procedures, access to relevant information and the ability to challenge decisions and practices.

92. States should seek to ensure that workers and other persons in employment or workplace contexts are not compelled, directly or indirectly, or subjected to undue pressure or incentives to undergo neurotechnological interventions, monitoring, assessment or enhancement, including as a condition for recruitment, hiring, continued employment, promotion, performance evaluation, remuneration, benefits, access to opportunities or other work-related advantages, except to the extent that any such measure is compliant with international human rights law and the relevant use or application is subject to appropriate safeguards, oversight and effective remedies.

93. States should seek to ensure that neurotechnologies and neurotechnology-related data and information used for legitimate purposes in employment and workplace contexts are not subsequently repurposed for unrelated performance evaluation, behavioral profiling, discriminatory treatment or other purposes incompatible with international human rights law, including through misuse, abuse or function creep.

94. States should pay particular attention to the long-term societal implications of workplace neurotechnological surveillance, behavioral monitoring, cognitive profiling and enhancement practices, including risks relating to inequality, exclusion, dependency, concentration of economic power and erosion of human autonomy.

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95. States should afford particular safeguards and protection to persons in situations of vulnerability or dependency in employment and workplace contexts, including children, older persons, persons with disabilities, persons with physical or mental health conditions, migrant workers, domestic workers, platform or gig-economy workers, military or security personnel, and persons exposed to structural inequalities or significant imbalances of power.

    States should pay particular attention to accessibility, inclusion, respect for neurodiversity, autonomy, equality and non-discrimination, meaningful participation and consultation of affected persons, and the effective enjoyment of the highest attainable standard of physical and mental health.

96. Workers and their representatives should be meaningfully informed, consulted and, where appropriate, involved in decision-making processes regarding the introduction and use of neurotechnologies in employment and workplace contexts, particularly where such technologies may significantly affect their rights, interests, working conditions, health, safety or well-being.

Criminal Justice and Deprivation of Liberty Settings

97. States should exercise particular caution regarding the development, deployment and use of neurotechnologies in criminal justice and law enforcement contexts, as well as in prisons, detention facilities, psychiatric institutions, immigration detention centers or any other custodial or institutional settings involving *de jure* or *de facto* deprivation of liberty by public authorities or private entities, in light of the coercive nature of such contexts and the significant imbalances of power and dependency relationships that may arise.

98. Where neurotechnologies are deployed or used in criminal justice, law enforcement or deprivation of liberty contexts for legitimate purposes, including safety, health, the prevention of harm or the protection of affected persons, detainees, staff or third parties, States should seek to ensure that such deployment or use is consistent with international human rights law and subject to appropriate safeguards, oversight and effective remedies.

    States should establish appropriate legal, institutional, policy and oversight mechanisms governing the development, deployment and use of neurotechnologies in such contexts, including, as appropriate, independent supervision, human rights impact assessments, testing and validation measures to identify and mitigate algorithmic bias, transparency requirements, accessible complaints procedures and effective review and redress mechanisms.

99. States should invest in specialized training for law enforcement officials, prosecutors, judges and defense counsel in relation to neurotechnologies, including risk evaluation and responsible decision-making.

100. States should regulate the use of neurotechnologies for investigative, surveillance or evidentiary purposes with a view to preventing arbitrary, abusive or coercive practices, discrimination or other adverse human rights impacts. Such uses should be subject to appropriate safeguards, including, where appropriate, prior authorization and oversight by an independent judicial authority.

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101. States should seek to prevent and address the collection, extraction, disclosure or use of neurotechnology-related data and information for criminal justice, law enforcement or intelligence purposes where such practices would undermine human dignity, freedom of thought, freedom of opinion and expression, physical and mental integrity, privacy, the right against self-incrimination, fair trial guarantees or other applicable human rights. States should prohibit such practices where they are incompatible with international human rights law.

States should exercise particular caution and apply heightened safeguards to the compelled extraction, disclosure or use of especially intimate cognitive content, including thoughts, memories, beliefs, intentions, emotions or other highly sensitive cognitive or emotional information. Such practices should be prohibited. If, in some circumstances, they may nevertheless be carried out consistently with international human rights law, they should be subject to prior authorization and oversight by an independent judicial authority and to heightened safeguards.

102. States should take particular care to ensure that, where consent is relied upon in relation to the use or application of neurotechnologies in criminal justice, law enforcement or deprivation of liberty contexts, particularly in situations involving institutionalization, dependency relationships or significant imbalances of power, such consent possesses the requisite characteristics and its validity is subject to rigorous scrutiny. Where appropriate, affected persons should have access to independent legal advice.

Even where consent possesses the requisite characteristics, it may, in some circumstances, not in itself be sufficient to justify a particular use or application.

Certain practices involving especially intimate cognitive content may remain incompatible with human dignity, freedom of thought or other applicable human rights rules even where valid consent has been given.

103. States should subject the use of neurotechnologies intended to assess or infer truthfulness, credibility, memory, recognition, intention, emotion, knowledge, deception or future behavior in criminal justice or law enforcement contexts, including through neural lie detection, concealed information tests or similar techniques, to strict regulation, heightened scrutiny and appropriate safeguards.

Such technologies should not be regarded as inherently reliable, objective or infallible. Where neurotechnology-related evidence is proposed for admission in judicial proceedings, States should seek to ensure rigorous scientific review and validation, including independent assessment, as appropriate, of its safety, validity, reliability, reproducibility, known limitations and suitability for the specific context and population in which it is applied, as well as appropriate procedural safeguards, including the ability to challenge its admission and use. States should also seek to ensure transparency concerning its scientific basis, limitations and reliability.

Compelled uses of such neurotechnologies should be prohibited. If, in some circumstances, they may nevertheless be carried out consistently with international human rights law, they should be subject to strict limitations and heightened safeguards, in light of the risks they may pose to freedom of thought, mental integrity, the right against self-incrimination and fair trial guarantees.

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104. States should seek to ensure that the use of neurotechnologies and neurotechnology-related data and information in criminal proceedings remains consistent with recognition as a person before the law, equality before the law, the right to liberty and security of person, including freedom from arbitrary detention, and with applicable fair trial and due process guarantees, including judicial independence, the presumption of innocence, the right against self-incrimination, equality of arms, the right to challenge evidence, access to relevant expertise, independent judicial assessment and effective judicial review.

States should seek to ensure that neural data and other particularly sensitive neurotechnology-related data and information collected or generated in consumer contexts are not introduced or relied upon as evidence in legal proceedings without due regard to their particularly sensitive character and appropriate safeguards, including in counterterrorism and national security contexts.

105. States should seek to ensure that neurotechnology-related data and information or evidence obtained through practices incompatible with international human rights law are not relied upon in criminal proceedings where such reliance would be incompatible with applicable fair trial guarantees or prohibited by international human rights law, including where such evidence has been obtained through violations of freedom of thought or the right against self-incrimination.

States should prohibit the use in criminal proceedings of statements, confessions or other evidence obtained by means of torture, in accordance with applicable international human rights law.

106. States should seek to prevent and address discrimination, stigmatization, arbitrariness, profiling based on race, social status or other protected characteristics, or other adverse human rights impacts arising from the use of neurotechnologies for predictive policing, behavioral prediction or risk assessment purposes, including those arising from biased neurotechnology-related data and information, probabilistic inferences or unequal social and structural conditions.

States should prohibit such uses where they cannot be carried out in a manner consistent with international human rights law.

107. States should exercise particular caution regarding the use of neurotechnologies in custodial or other institutional settings of deprivation of liberty where they are capable of influencing cognition, emotion, behavior, decision-making or the exercise of independent judgment.

States should seek to prevent and address the use of neurotechnologies in such settings for purposes of coercion, punishment or behavioral control.

States should prohibit their use as a means of torture or other cruel, inhuman or degrading treatment or punishment, or for purposes of subjugation or dehumanization, as well as any other use incompatible with human dignity or international human rights law.

108. States should seek to ensure that neurotechnologies employed for protective, therapeutic, assistive, preventive, safety-related or other legitimate purposes are not

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repurposed for coercive, punitive, disciplinary, discriminatory, manipulative or otherwise incompatible purposes, including through misuse, abuse or function creep.

109. States should distinguish between legitimate therapeutic or assistive uses of neurotechnologies in settings of *de jure* or *de facto* deprivation of liberty by public authorities or private entities, including uses relating to healthcare, medical or functional rehabilitation, accessibility, communication or other forms of support, and uses primarily intended to enforce compliance, suppress behavior, facilitate control or otherwise interfere with autonomy, mental integrity or freedom of thought.

States should also distinguish such therapeutic or assistive uses from preventive, protective or safety-related uses of neurotechnologies in such settings.

110. States should seek to ensure that persons held in situations of *de jure* or *de facto* deprivation of liberty by public authorities or private entities are not compelled, directly or indirectly, to undergo neurotechnological interventions or monitoring except to the extent that such compulsion is compliant with international human rights law and the relevant use or application is subject to appropriate safeguards, oversight and effective remedies.

States should exercise particular caution where neurotechnological interventions or monitoring are imposed as a condition for the enjoyment or exercise of rights or access to services, medical care, parole, reduced restrictions or other benefits.

111. States should prohibit the use of neurotechnologies in relation to children in criminal justice, law enforcement or custodial contexts, including in situations of *de jure* or *de facto* deprivation of liberty by public authorities or private entities, where, taking into account the best interests of the child, the nature and severity of the risks to their human rights cannot be adequately addressed through appropriate safeguards. Any use that is not prohibited should be subject to heightened scrutiny, appropriate safeguards, oversight and effective remedies.

In all actions concerning children, the best interests of the child should be a guiding principle and a primary consideration. States should pay particular attention to children's age, evolving capacities, vulnerability, neural and psychological development, and heightened exposure to coercion, dependency relationships, manipulation and long-term adverse effects.

Military, Security and Conflict-Related Uses

112. States should exercise particular caution regarding the development, deployment and use of neurotechnologies in military, security or conflict-related contexts, particularly where these are characterized by hierarchical command structures, in light of the significant imbalances of power, dependency relationships and constraints on the ability to provide consent possessing the requisite characteristics that may arise in such contexts.

Particular caution is also warranted regarding practices involving neurotechnologies that affect cognition, emotions or behavior, including through behavioral conditioning, emotional modulation or cognitive interference.

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113. States should prohibit the use of neurotechnologies for destabilizing purposes or in ways that undermine peace or security in a manner incompatible with international law.

States should prohibit the use of neurotechnologies as a means of aggression enhancement, suppression of empathy or dehumanization, or for the purpose of impairing or overriding individuals' ability to exercise independent judgment.

114. States should subject the development, acquisition, deployment and use of neurotechnologies intended or capable of being used as weapons, or as means or methods of warfare or security operations, to appropriate legal review and seek to ensure their conformity with applicable international human rights law and, where applicable, international humanitarian law and international criminal law.

115. Where the development, deployment or use of neurotechnologies in military, security or conflict-related contexts is permissible under international human rights law and, where applicable, international humanitarian law, including for purposes relating to safety, health, operational protection, the prevention of harm, the protection of military or security personnel, civilians or third parties, or other legitimate purposes, States should seek to ensure that such development, deployment or use is subject to appropriate safeguards, oversight and effective remedies.

116. States should seek to ensure that military or security personnel are not compelled, directly or indirectly, to use neurotechnologies or otherwise be subject to their application, except to the extent that such compulsion is compliant with international human rights law and the relevant use or application is subject to appropriate safeguards, oversight and effective remedies.

Surveillance, Monitoring and Tracking

117. States should exercise particular caution regarding the development, deployment and use of neurotechnologies capable of enabling surveillance, monitoring, tracking, inference, analysis or interpretation of mental or neural processes, including thoughts, beliefs, opinions, emotions, intentions, attention, cognitive states or other forms of especially intimate cognitive or emotional information. Particular caution is warranted regarding practices involving pervasive or continuous monitoring of mental or neural processes, intrusive monitoring or tracking, or chilling effects on thought, expression or behavior.

118. States should prohibit neurotechnological surveillance, monitoring or tracking practices that are incompatible with international human rights law.

119. Where the use of neurotechnologies for surveillance purposes is permissible under international human rights law, States should establish appropriate legal, institutional, policy and oversight mechanisms, including, where appropriate, prior authorization by an independent judicial authority, transparency requirements, independent supervision, human rights impact assessments, accessible complaints procedures and effective review and redress mechanisms.

120. States should pay particular attention to the heightened sensitivity of mental and cognitive processes and to the risks arising from the aggregation, inference, predictive use

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or long-term analysis of neurotechnology-related data and information, particularly where such practices enable or facilitate surveillance, monitoring or tracking of individuals.

Civic Participation and the Rule of Law, and Peaceful Uses

121. States should seek to ensure that neurotechnologies may, where appropriate and consistent with international human rights law, contribute to accessibility, inclusion, equality and the meaningful and effective participation of all persons in civic, cultural, social, public and political life, including persons who might otherwise face exclusion, marginalization or barriers to participation.

122. States should not deploy or use neurotechnologies in ways capable of impairing free and informed civic participation, political agency, public discourse or compromising institutions and processes essential to the rule of law, including judicial independence. Where such deployment or use interferes with qualified human rights, it should remain consistent with international human rights law and be subject to appropriate safeguards, oversight and effective remedies.

123. States should seek to ensure that neurotechnologies are not deployed or used to impair free and informed civic participation, political agency, public discourse or compromise institutions and processes essential to the rule of law, including judicial independence. To that end, States should take, as appropriate, measures to prevent and stop such practices, investigate alleged human rights violations, ensure accountability and provide access to effective remedies.

Such measures should also seek to prevent and address the use of neurotechnologies to manipulate or otherwise influence or alter individuals' mental processes, emotions, decision-making, personality or sense of identity in ways incompatible with their autonomy, freedom of thought or opinion, mental integrity or other applicable human rights.

124. States should seek to ensure that neurotechnologies are not deployed or used in connection with peaceful assemblies, demonstrations or other forms of collective public expression in ways incompatible with the rights to peaceful assembly and freedom of expression or other applicable human rights.

Where neurotechnologies are deployed or used for law enforcement, public-order or crowd-control purposes in such contexts, their deployment and use should comply with international human rights law and be subject to appropriate safeguards, oversight and effective remedies.

125. States should promote the peaceful, safe and human rights-consistent governance and use of neurotechnologies across their lifecycle, including through appropriate legal, regulatory, institutional and oversight frameworks, international cooperation, interdisciplinary research, public awareness and human rights-based public education aimed at strengthening neurotechnology literacy.

126. States should promote the development and use of human rights-compliant neurotechnologies with a view to enabling individuals to enjoy the benefits of scientific

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progress and its applications. In doing so, the interests and welfare of individuals should not be overridden solely by considerations of scientific or societal interests.

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